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Alpilla 600 MW Combined Cycle Power Plant

Natural Gas Quality and Supply Conditions

Document number	ALP-OWN-FUL-001
Revision	0
Date	2026-09-08
Status	Issued for tender / contract
Confidentiality	Restricted - Contract ALP-EPC-001 parties only

	Prepared	Checked	Approved
Function	Owner's Engineer (discipline lead)	Owner's Engineer (project engineer)	Owner's Project Director
Signature / date	signed 2026-09-08	signed 2026-09-08	signed 2026-09-08

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CASE STUDY - all organisations, data and signatures are fictional.

1. Basis

Gas transmission network operator's quality data for the regional high-pressure line (2022–2025, on-line chromatograph at the nearest compressor station, 35 km upstream), the operator's network entry specification, and the Owner's gas transport agreement. Reference state for volumes: 15 °C, 1.01325 bar ("Sm³"); energies on LHV unless stated.

2. Composition (mol %)

Component	Minimum	Typical (design)	Maximum
Methane CH ₄	85.0	93.0	98.5
Ethane C ₂ H ₆	0.5	3.5	7.0
Propane C ₃ H ₈	0.05	0.80	2.5
i-Butane	0.00	0.15	0.50
n-Butane	0.00	0.15	0.50
Pentanes and heavier (C ₅ +)	0.00	0.05	0.20
Nitrogen N ₂	0.30	1.55	4.0
Carbon dioxide CO ₂	0.00	0.80	2.5
Oxygen O ₂	0.000	0.001	0.02
Hydrogen H ₂	0.0	< 0.1	see 5
Helium	0.00	0.01	0.05

3. Physical properties

Property	Unit	Minimum	Typical	Maximum
LHV (mass)	MJ/kg	44.2	48.4	49.8
LHV (volume)	MJ/Sm ³	31.4	34.4	37.9
GCV (volume)	MJ/Sm ³	35.0	38.2	42.0
Wobbe index (GCV)	MJ/Sm ³	47.2	50.2	51.8
Relative density	-	0.560	0.600	0.690
Rate of change of Wobbe index	%/min			1.0
Methane number	-	72	88	97

Note: the lower end of the LHV range (44.2 MJ/kg) occurs with high N₂/CO₂ gas from one entry point; it has occurred for 3 % of hours.

4. Contaminants and trace components

Parameter	Unit	Network limit	Measured typical	Measured max
Total sulphur (incl. H ₂ S, COS, mercaptans)	mg S/Sm ³	30	4	17
H ₂ S + COS (as S)	mg/Sm ³	5	0.8	3.1
Mercaptan sulphur	mg/Sm ³	6	0.5	2.2
Mercury	µg/Sm ³	1	< 0.01	0.03
Water dew point	°C at 70 barg	≤ -8	-14	-9
Hydrocarbon dew point	°C (1–70 barg)	≤ -2	-9	-4
Particulates (> 10 µm)	mg/Sm ³	1	n.m.	n.m.
Sodium + potassium (alkali metals)	mg/Sm ³	not specified	< 0.005 (d.l.)	< 0.005
Lead	mg/Sm ³	not specified	< 0.005 (d.l.)	< 0.005
Vanadium	mg/Sm ³	not specified	< 0.005 (d.l.)	< 0.005
Calcium	mg/Sm ³	not specified	< 0.01 (d.l.)	< 0.01
Chlorides	mg/Sm ³	not specified	< 0.05 (d.l.)	< 0.05
Compressor oil carry-over	mg/Sm ³	not specified	not measured	not measured
Odorant	-	transmission gas is not odorised		

d.l. = detection limit of the laboratory method; n.m. = not measured. The network operator does not guarantee trace metal, chloride or compressor oil content. The Owner requires the Contractor's fuel gas conditioning to protect the gas turbine against liquid carry-over and particulates under all network conditions.

5. Hydrogen blending

The network operator has announced hydrogen blending pilots of up to 5 vol % from 2030 and up to 10 vol % after 2035 (not firm). The plant shall be capable of 10 vol % H₂ (contract ER-03.02); the Owner accepts reduced emissions guarantees above 5 vol % if justified by the OEM.

6. Supply pressure, temperature and capacity at the plant tie-in (TP-G1)

Parameter	Value
Normal pressure	55–65 barg
Minimum guaranteed pressure	45 barg
Exceptional minimum (winter peak, max 72 h/year, notified 24 h in advance)	40 barg
Maximum operating pressure / MAOP	70 barg / 75 barg
Pressure rate of change	≤ 1.0 bar/min
Temperature	5–25 °C
Booked capacity	115,000 Sm ³ /h firm; 130,000 Sm ³ /h interruptible
Fiscal metering	by the Contractor per network operator standards; custody transfer at the metering station outlet

The network operator requires a slam-shut and relief protection at the station inlet and approval of the metering design.